

How to Reduce Power Consumption in Battery-Powered Devices

Battery technology lags far behind the technology it powers. As a result, the sole area in which portable devices are stagnant is their battery life. It's time designers focused on extracting more life from the same amount of charge. This article tells you how to do it



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Over the last few years, we saw numerous new battery technologies being proposed by different research institutes. There was news of flexible Li-ion batteries and even batteries made from graphene. However, no one is really sure how long it would take to get these new technologies into commercial form and therefore there is no point in waiting for them at this point of time.

If you cannot increase the amount of charge available for your device, the next best thing is to increase the amount of work that it can do with

the same amount of charge. That is why we decided to check with the top engineers in the industry how they have been enhancing battery life in their devices.

How to save power

There are umpteen areas that you can focus on to save power in a system. The major ones are discussed below.

Select the right component. While working on a project that focuses on operating with very low power consumption, it is of paramount importance to choose components that improve the power system performance within the constraints of the project's

current budget. One way to do this is to utilise power saving techniques on top of your low-power components.

Brian Chu, senior application engineer, Microchip Technology, says, "In addition to selecting low-supply-current devices, it is recommended that designers take advantage of various power-saving techniques such as load-based multi-mode power-conversion design, input-to-output bypass design, standby and the like while maintaining the performance level."

Fortunately, all the latest components available play a part in helping engineers design more efficient circuits. The latest power switches offer